In [1]:
```
import pandas as pd
```

```
/home/student/.local/lib/python3.9/site-packages/pandas/core/compu
tation/expressions.py:21: UserWarning: Pandas requires version '2.
8.4' or newer of 'numexpr' (version '2.8.1' currently installed).
  from pandas.core.computation.check import NUMEXPR_INSTALLED
/home/student/.local/lib/python3.9/site-packages/pandas/core/array
s/masked.py:61: UserWarning: Pandas requires version '1.3.6' or ne
wer of 'bottleneck' (version '1.3.4' currently installed).
  from pandas.core import (
```

In [69]:
```
df=pd.read_csv("students.csv")
```

In [70]:
```
df.head(10)
```

Out[70]:

	StudentID	Name	Gender	Age	Course	Marks	Attendence	City
0	101	Asha	F	20	BCA	78	85	Ahmedabad
1	102	Ravi	M	21	Bsc	65	70	Surat
2	103	Neha	F	19	BCA	88	90	Vadodara
3	104	Aman	M	22	Bcom	55	60	Rajkot
4	105	Pooja	F	20	Bsc	92	95	Ahmedabad
5	106	Astha	F	20	BCA	85	85	Ahmedabad
6	107	Rakesh	M	21	Bsc	74	69	Surat
7	108	Nisha	F	19	BCA	96	74	Vadodara
8	109	Arpit	M	22	Bcom	85	45	Rajkot
9	110	Khushi	F	20	Bsc	87	88	Ahmedabad

In [71]:
```
df.tail(5)
```

Out[71]:

	StudentID	Name	Gender	Age	Course	Marks	Attendence	City
5	106	Astha	F	20	BCA	85	85	Ahmedabad
6	107	Rakesh	M	21	Bsc	74	69	Surat
7	108	Nisha	F	19	BCA	96	74	Vadodara
8	109	Arpit	M	22	Bcom	85	45	Rajkot
9	110	Khushi	F	20	Bsc	87	88	Ahmedabad

In [72]:
```
df.columns
```

Out[72]:
```
Index(['StudentID', 'Name', 'Gender', 'Age', 'Course', 'Marks', 'A
ttendence',
       'City'],
      dtype='object')
```

In [75]:
```
df.drop("Attendence",axis=1,inplace=True)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Input In [75], in <cell line: 1>()
----> 1 df.drop("Attendence",axis=1,inplace=True)

File ~/.local/lib/python3.9/site-packages/pandas/core/frame.py:5588, in DataFrame.drop(self, labels, axis, index, columns, level, inplace, errors)
   5440 def drop(
   5441     self,
   5442     labels: IndexLabel | None = None,
   (...)
   5449     errors: IgnoreRaise = "raise",
   5450 ) -> DataFrame | None:
   5451     """
   5452     Drop specified labels from rows or columns.
   5453
   (...)
   5586             weight  1.0     0.8
   5587     """
-> 5588     return super().drop(
   5589         labels=labels,
   5590         axis=axis,
   5591         index=index,
   5592         columns=columns,
   5593         level=level,
   5594         inplace=inplace,
   5595         errors=errors,
   5596     )

File ~/.local/lib/python3.9/site-packages/pandas/core/generic.py:4807, in NDFrame.drop(self, labels, axis, index, columns, level, inplace, errors)
   4805 for axis, labels in axes.items():
   4806     if labels is not None:
-> 4807         obj = obj._drop_axis(labels, axis, level=level, errors=errors)
   4809 if inplace:
   4810     self._update_inplace(obj)

File ~/.local/lib/python3.9/site-packages/pandas/core/generic.py:4849, in NDFrame._drop_axis(self, labels, axis, level, errors, only_slice)
   4847         new_axis = axis.drop(labels, level=level, errors=errors)
   4848     else:
-> 4849         new_axis = axis.drop(labels, errors=errors)
   4850     indexer = axis.get_indexer(new_axis)
   4852 # Case for non-unique axis
   4853 else:

File ~/.local/lib/python3.9/site-packages/pandas/core/indexes/base.py:7136, in Index.drop(self, labels, errors)
   7134 if mask.any():
   7135     if errors != "ignore":
-> 7136         raise KeyError(f"{labels[mask].tolist()} not found in axis")
   7137     indexer = indexer[~mask]
   7138 return self.delete(indexer)
```

```
KeyError: "['Attendence'] not found in axis"
```

In [76]:
```
new_df=df[["Name","Course","Marks"]]
new_df
```

Out[76]:

	Name	Course	Marks
0	Asha	BCA	78
1	Ravi	Bsc	65
2	Neha	BCA	88
3	Aman	Bcom	55
4	Pooja	Bsc	92
5	Astha	BCA	85
6	Rakesh	Bsc	74
7	Nisha	BCA	96
8	Arpit	Bcom	85
9	Khushi	Bsc	87

In [77]:
```
df[df["Marks"]>80]
```

Out[77]:

	StudentID	Name	Gender	Age	Course	Marks	City
2	103	Neha	F	19	BCA	88	Vadodara
4	105	Pooja	F	20	Bsc	92	Ahmedabad
5	106	Astha	F	20	BCA	85	Ahmedabad
7	108	Nisha	F	19	BCA	96	Vadodara
8	109	Arpit	M	22	Bcom	85	Rajkot
9	110	Khushi	F	20	Bsc	87	Ahmedabad

In [78]:
```
df["Course"].value_counts()
```

Out[78]:
```
Course
BCA     4
Bsc     4
Bcom    2
Name: count, dtype: int64
```

In [79]:
```
df.isnull().sum()
```

Out[79]:
```
StudentID    0
Name         0
Gender       0
Age          0
Course       0
Marks        0
City         0
dtype: int64
```

In [80]:
```
df.describe()
```

Out[80]:

	StudentID	Age	Marks
count	10.00000	10.000000	10.000000
mean	105.50000	20.400000	80.500000
std	3.02765	1.074968	12.695143
min	101.00000	19.000000	55.000000
25%	103.25000	20.000000	75.000000
50%	105.50000	20.000000	85.000000
75%	107.75000	21.000000	87.750000
max	110.00000	22.000000	96.000000

In [81]:
```
df.sort_values(by="Marks",ascending=False)
```

Out[81]:

	StudentID	Name	Gender	Age	Course	Marks	City
7	108	Nisha	F	19	BCA	96	Vadodara
4	105	Pooja	F	20	Bsc	92	Ahmedabad
2	103	Neha	F	19	BCA	88	Vadodara
9	110	Khushi	F	20	Bsc	87	Ahmedabad
5	106	Astha	F	20	BCA	85	Ahmedabad
8	109	Arpit	M	22	Bcom	85	Rajkot
0	101	Asha	F	20	BCA	78	Ahmedabad
6	107	Rakesh	M	21	Bsc	74	Surat
1	102	Ravi	M	21	Bsc	65	Surat
3	104	Aman	M	22	Bcom	55	Rajkot

In [82]:
```
import numpy as np
```

In [83]:
```
marks=np.array([45,67,89,90,56,72,88,95,60,70])
```

In [84]:
```
marks.max(),marks.min(),marks.mean()
```

Out[84]:
```
(95, 45, 73.2)
```

In [85]:
```
marks.reshape(2,5)
```

Out[85]:
```
array([[45, 67, 89, 90, 56],
       [72, 88, 95, 60, 70]])
```

In [87]:
```
marks[marks>70]
```

Out[87]:
```
array([89, 90, 72, 88, 95])
```

In [88]:
```
marks+5
```

Out[88]:
```
array([ 50,  72,  94,  95,  61,  77,  93, 100,  65,  75])
```

In [89]:
```
np.sort(marks)
```

Out[89]:
```
array([45, 56, 60, 67, 70, 72, 88, 89, 90, 95])
```

In [90]:
```
np.std(marks),np.var(marks)
```

Out[90]:
```
(15.942396306703708, 254.16000000000003)
```

In [91]:
```
np.arange(3,46,3)
```

Out[91]:
```
array([ 3,  6,  9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 4
5])
```

In [92]:
```
np.random.randint(10,100,10)
```

Out[92]:
```
array([50, 32, 26, 89, 15, 58, 77, 79, 24, 69])
```

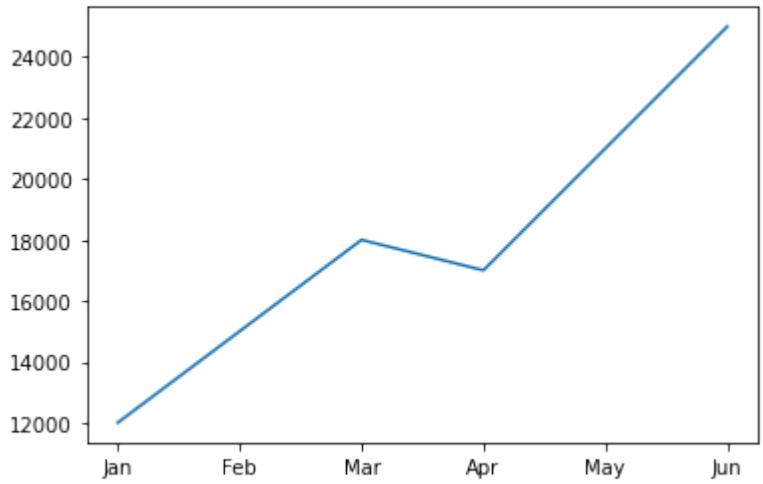
In [93]:
```
np.where(marks<60,60,marks)
```

Out[93]:
```
array([60, 67, 89, 90, 60, 72, 88, 95, 60, 70])
```

In [94]:
```
import matplotlib.pyplot as plt
```

In [95]:
```
months=["Jan","Feb","Mar","Apr","May","Jun"]
sales=[12000,15000,18000,17000,21000,25000]
```

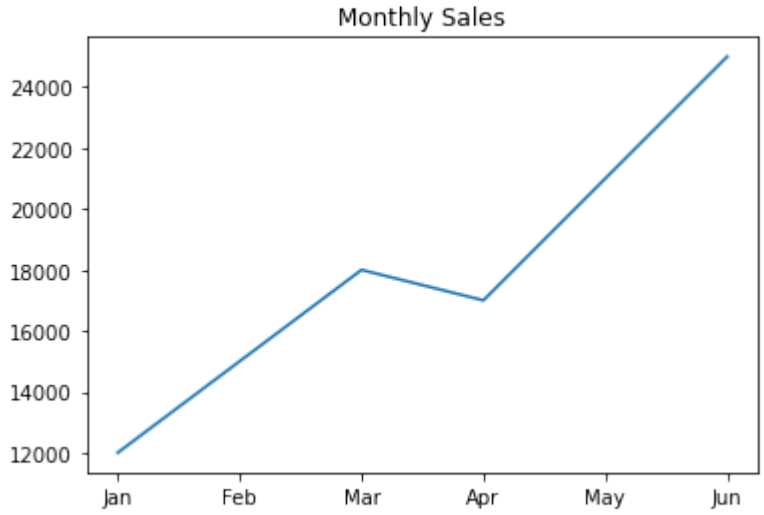
In [96]:
```
plt.plot(months,sales)
plt.show()
```



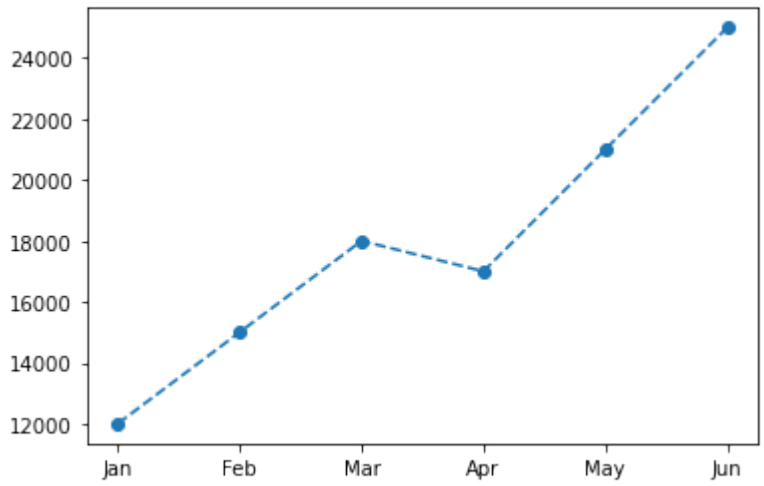
In [97]:
```
plt.plot(months,sales)
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Input In [97], in <cell line: 3>()
      1 plt.plot(months,sales)
      2 plt.title("Monthly Sales")
----> 3 plt.xlabel("Month")
      4 plt.ylabel("Sales")
      5 plt.show()

TypeError: 'str' object is not callable
```

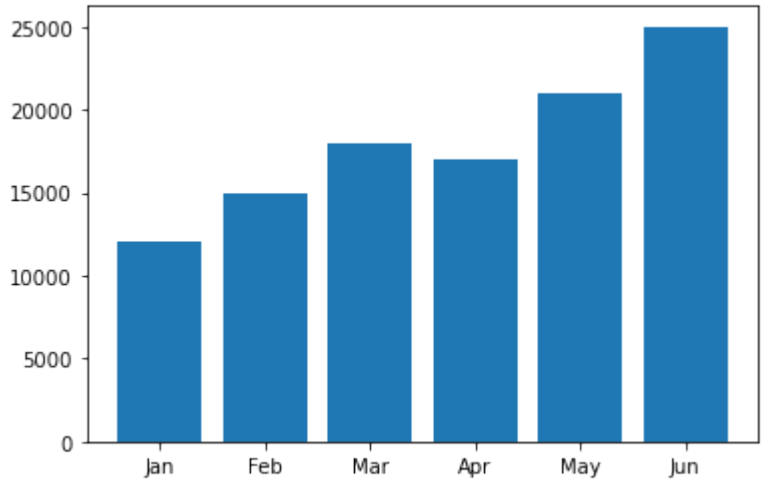


In [98]:
```
plt.plot(months,sales,linestyle="--",marker="o")
plt.show()
```

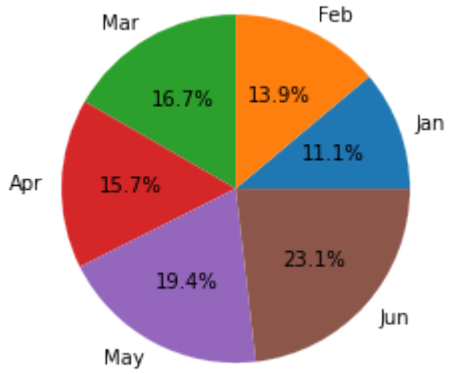


In [99]:
```
plt.bar(months,sales)
```

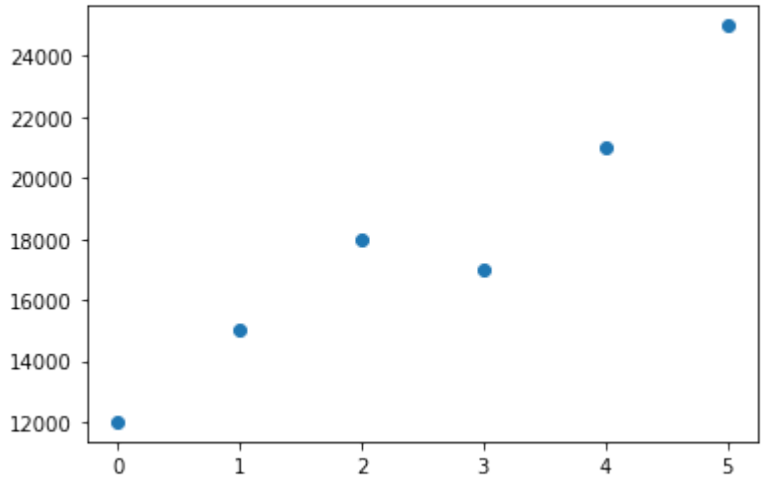
Out[99]: <BarContainer object of 6 artists>



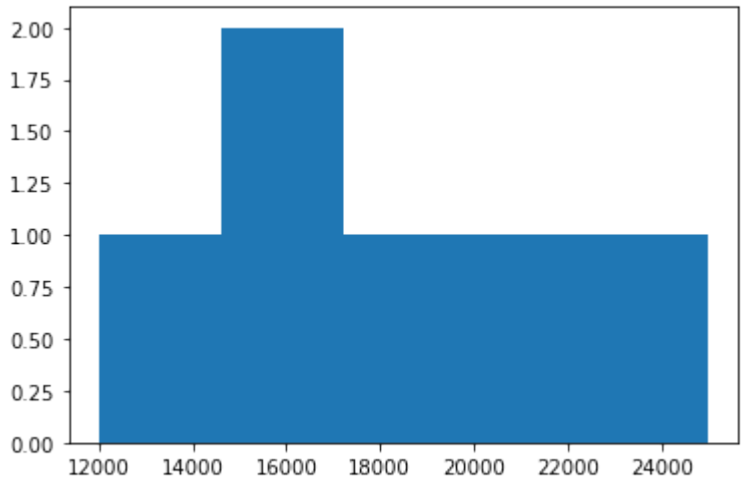
In [100]:
```
plt.pie(sales,labels=months,autopct="%1.1f%%")
plt.show()
```



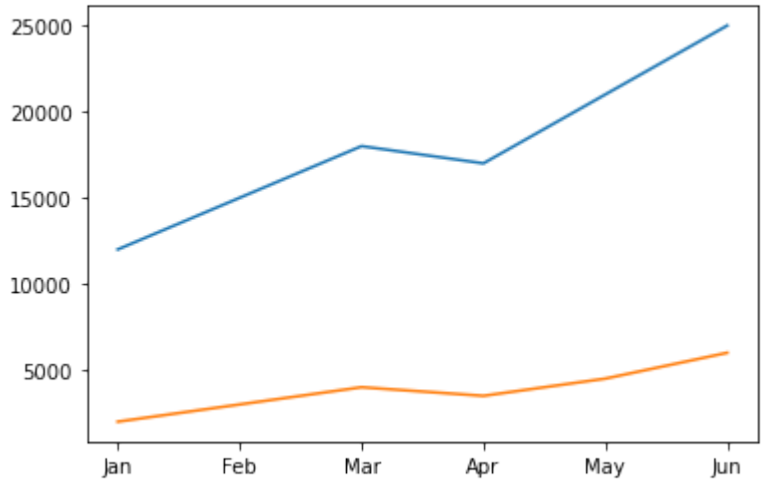
In [101]:
```
plt.scatter(range(len(months)),sales)
plt.show()
```



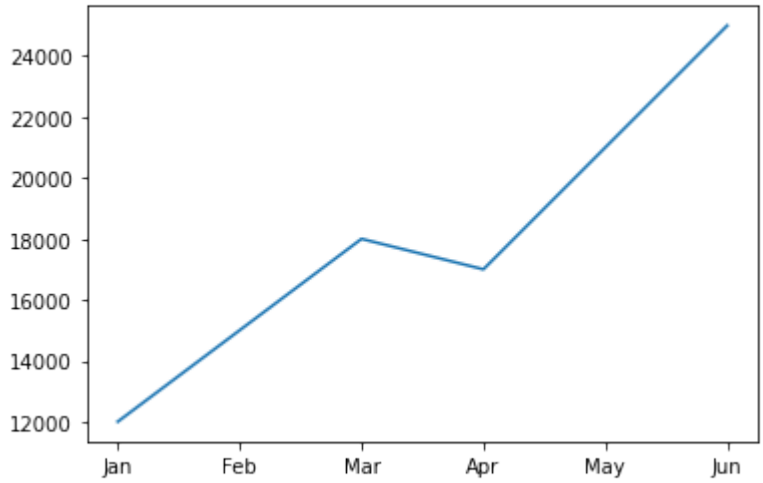
In [102]:
```
plt.hist(sales,bins=5)
plt.show()
```



In [103]:
```
profit=[2000,3000,4000,3500,4500,6000]
plt.plot(months,sales)
plt.plot(months,profit)
plt.show()
```



In [104]:
```
plt.plot(months,sales)
plt.savefig("sales_chart.png")
plt.show()
```



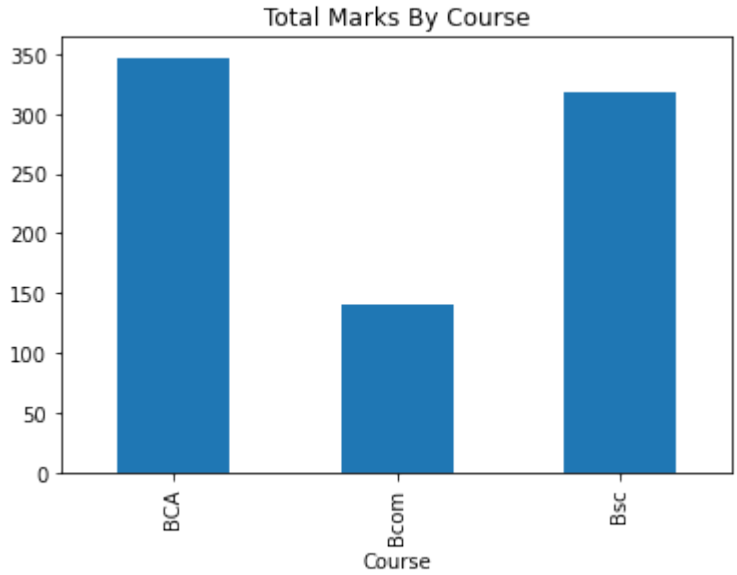
In [105]:
```
df=pd.read_csv("students.csv")
print(type(df))
```

```
<class 'pandas.core.frame.DataFrame'>
```

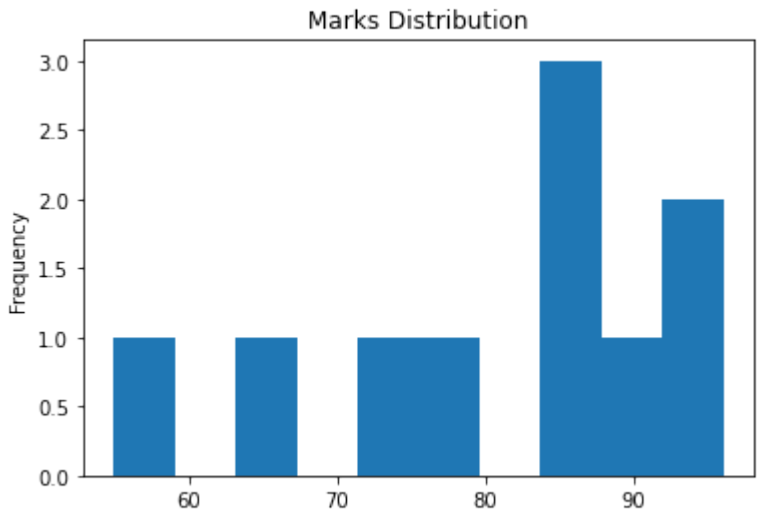
In [106]:
```
grouped=df.groupby("Course")["Marks"].sum()
print(grouped)
```

```
Course
BCA     347
Bcom    140
Bsc     318
Name: Marks, dtype: int64
```

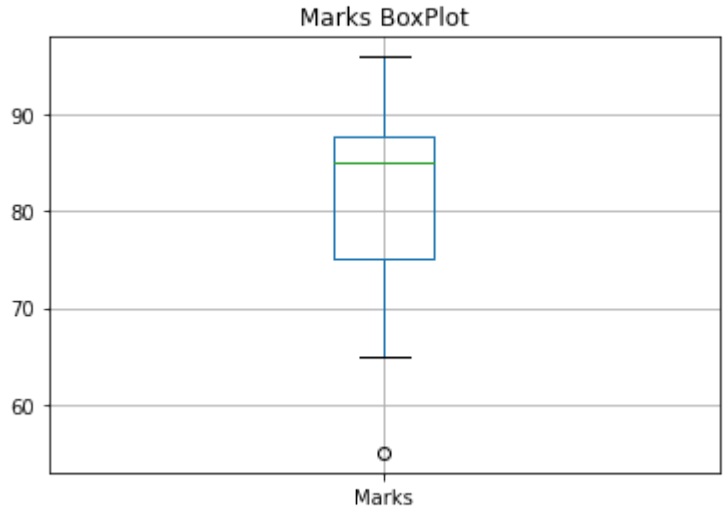
In [107]:
```
grouped.plot(kind="bar")
plt.title("Total Marks By Course")
plt.xlabel=("Course")
plt.ylabel=("Marks")
plt.show()
```



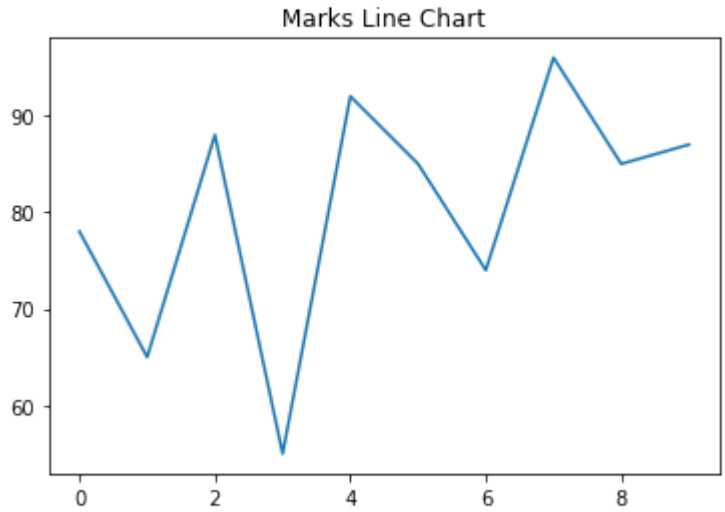
In [108]:
```
df["Marks"].plot(kind="hist")
plt.title("Marks Distribution")
plt.show()
```

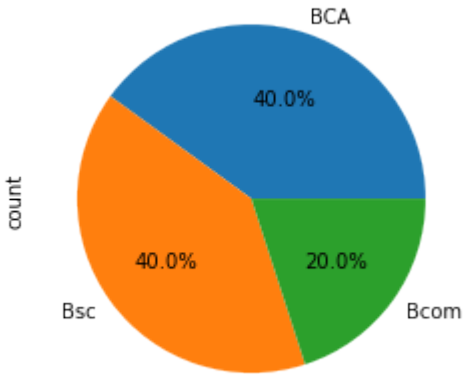


In [109]:
```
df.boxplot(column="Marks")
plt.title("Marks BoxPlot")
plt.show()
```



In [110]:
```
df["Marks"].plot(kind="line")
plt.title("Marks Line Chart")
plt.show()
```



In [111]:
```
df["Course"].value_counts().plot(
kind="pie",
autopct="%1.1f%%"
)
plt.ylabel("")
plt.show()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Input In [111], in <cell line: 5>()
      1 df["Course"].value_counts().plot(
      2 kind="pie",
      3 autopct="%1.1f%%"
      4 )
----> 5 plt.ylabel("")
      6 plt.show()

TypeError: 'str' object is not callable
```